



Gridlock Hackathon 2.0 - Round 2

Hosted By - Flipkart

PROTOTYPE ROUND 2

Jun 15 – Jun 21, 2026

10:00 PM – 11:59 PM · Asia/Kolkata

🌐 Online · 👥 Team size: 1 · 📈 1.6K registrations



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GRIDLOCK HACKATHON

solve . traffic

CONGRATULATIONS!

Your team has successfully cleared the **Round 1 ML Challenge** and has been shortlisted for **Round 2 – the Prototype Phase** of Gridlock Hackathon 2.0.

This is a significant achievement, and we are excited to see your solution evolve into a working prototype. The **Prototype Phase** officially begins today!

Review the available problem statements, select the one you wish to solve, and **start building your prototype**.

Make sure to submit your prototype **before the deadline**.

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We wish you the very best and look forward to seeing your **innovative solutions** in action!

Your Journey

The hackathon is structured across three phases, progressing from an online ML prototype and culminating in an onsite finale at Flipkart HQ.

View Options

Now, your Prototype Phase journey begins. Select your problem statement, start building your solution, and bring your ideas to life.

PROTOTYPE DEVELOPMENT

Build Something Real

Shortlisted teams move into development on real Bengaluru traffic challenges with localized data and partner resources.

1 Tackle real Bengaluru problems

Shortlisted teams move into development focused on real-world Bengaluru traffic challenges.

2 Access exclusive datasets

Teams get simulated datasets from HackerEarth to build prototypes.

3 Expert prototype review

Final prototypes are evaluated by an expert panel on feasibility, relevance, innovation and real-world impact.

 Online · HackerEarth

FLIPKART HQ, BENGALURU

The Onsite Finale

Top 10 teams pitch live at Flipkart HQ in front of subject-matter experts and Bengaluru Traffic Police leadership.

1 Top 10 teams invited

Shortlisted finalists are invited to Flipkart HQ in Bengaluru. In-person attendance is mandatory to remain prize-eligible.

2 Present and demonstrate

Each team delivers a comprehensive final presentation and demonstrates a working prototype before an esteemed panel of subject-matter experts.

3 Expert evaluation

Judging covers solution robustness, innovation, prototype clarity, scalability, and real-world viability for Bengaluru's traffic.

4 Winners announced

Top 3 are felicitated by the Head of Bengaluru Traffic Police and Flipkart leadership. Top innovations get direct exposure to industry leaders.

 Flipkart HQ · Bengaluru

Themes



Poor Visibility on Parking-Induced Congestion

Operational Challenge

On-street illegal parking and spillover parking near commercial areas, metro stations, and events choke carriageways and intersections.

Why It's Hard Today

- Enforcement is patrol-based and reactive.
- No heatmap of parking violations vs. congestion impact.
- Difficult to prioritize enforcement zones.

Problem Statement Direction

How can AI-driven parking intelligence detect illegal parking hotspots and quantify their impact on traffic flow to enable targeted enforcement?

Dataset: [Link](#)



Event-Driven Congestion (Planned & Unplanned)

Operational Challenge

Political rallies, festivals, sports events, construction activities, and sudden gatherings create localized traffic breakdowns.

Why It's Hard Today

- Event impact is not quantified in advance.
- Resource deployment is experience-driven.
- No post-event learning system.

Problem Statement Direction

How can historical and real-time data be used to forecast event-related traffic impact and recommend optimal manpower, barricading, and diversion plans?

Dataset: [Link](#)



Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

Overview: You should submit a unique and innovative idea that addresses the stated challenge. Your idea should be practical, clearly explained, and capable of solving a real-world problem. You may present it in the form of a concept note, prototype proposal, or solution framework. With the increasing deployment of traffic surveillance cameras and automated monitoring systems, large volumes of traffic images are generated every day. Manual inspection of these images to identify traffic violations is labor-intensive, time-

consuming, and prone to inconsistencies. An intelligent system capable of automatically analyzing photographic evidence can significantly improve the efficiency and accuracy of traffic law enforcement.

The objective of this project is to develop a computer vision-based solution that can automatically process traffic images, detect vehicles and road users, identify traffic violations, classify the type of violation, and generate annotated evidence for further review. The system should be robust to varying environmental conditions, traffic densities, and image qualities while maintaining high detection accuracy and scalability.

Tasks

1. Image Preprocessing

- Enhance image quality and normalize inputs.
- Handle challenges such as low light, rain, shadows, and motion blur.

2. Vehicle and Road User Detection

- Detect and localize vehicles, riders, drivers, and pedestrians.
- Classify different vehicle categories.

3. Traffic Violation Detection

- Identify violations from images, including:
 - Helmet non-compliance
 - Seatbelt non-compliance
 - Triple riding
 - Wrong-side driving
 - Stop-line violation
 - Red-light violation
 - Illegal parking

4. Violation Classification

- Categorize detected violations into predefined classes.
- Assign confidence scores to predictions.

5. License Plate Recognition

- Detect number plates.
- Extract registration details using OCR techniques.

6. Evidence Generation

- Produce annotated images highlighting violations.
- Store violation metadata and timestamps.

7. Analytics and Reporting

- Generate violation statistics and trends.
- Provide searchable records and summary reports.

8. Performance Evaluation

- Evaluate using Accuracy, Precision, Recall, F1-score, and mAP.

- Assess computational efficiency and scalability.

Expected Outcome

A scalable AI-based traffic image analysis system capable of automatically identifying, classifying, and documenting traffic violations from photographic evidence, thereby reducing manual effort and improving the effectiveness of traffic monitoring and enforcement.

Prizes

Win incredible prizes and opportunities to showcase your innovation to the world. Total ₹500,000 in prizes to be won!

Main Prizes



First prize
₹225,000



Second prize
₹175,000



Third prize
₹100,000

Partners